

David Le Chan

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Education

- M.S. in Electrical and Computer Engineering (ECE)** Expected May 2027
Carnegie Mellon University (CMU) | Pittsburgh, PA
- GPA: /4.00
- B.S. in Electrical and Computer Engineering (ECE)** Graduated May 2026
Carnegie Mellon University (CMU) | Pittsburgh, PA
- GPA: 3.8/4.0; *University Honors, 6x College of Engineering Dean's List Recipient*

Work Experience

- Hardware Development Engineering Intern** May 2026 - August 2026
Annapurna Labs (Amazon AWS), Machine Learning Acceleration | Austin, TX
- Validated and deployed PCIe Gen5/Gen6 connectivity to scale machine learning acceleration systems
- Designed and executed qualification tests for switches, expanding high-speed rack-level test coverage
- Migrated tests to a Lua orchestration framework to enable continuous validation via AWS test infrastructure
- Identified mechanical defects impacting 100k+ units, driving resolution to meet deployment schedules
- FPGA & Electrical Engineering Intern** May 2025 - August 2025
KLA Corporation, Broadband Plasma Semiconductor Inspection | Milpitas, CA
- Upgraded image compression FPGA firmware, achieving a 6.7% throughput increase on flagship inspection tools
- Designed Verilog pipelines for image segmentation and reconstruction, verifying functionality via Questa
- Validated designs on Alveo PCIe cards, closing timing at 80MHz with 50% more parallel processing engines
- Undergraduate Research Assistant** May 2024 - Present
IO Harness Project, CMU ECE | Pittsburgh, PA
- Designing a standardized research-chip harness to minimize infrastructure rework; testing a 180nm prototype
- Authored and verified SystemVerilog RTL for UART, I2C, and SPI communication blocks
- Streamlining an eFPGA-generation pipeline to serve as a device under test (DUT)
- Power Electronics & Programming Intern** May 2023 - August 2023
Tau Motors | Redwood City, CA
- Independently designed and prototyped motor testbenches and PCBs, owning layout, assembly, and validation
- Built Python inventory management tools from scratch to accelerate hardware testing cycles

Leadership and Projects

- Hybrid FFT/NTT Accelerator SoC** January 2026 - Present
- Designed a reconfigurable accelerator, executing the entire RTL-to-GDSII flow for tapeout in a 28nm node
- Integrated a RISC-V core in SpinalHDL to benchmark accelerator performance against software baselines
- Built a custom cocotb framework to validate the SoC across all design hierarchies and stages
- Head Teaching Assistant (TA), Introduction to ECE (18-100)** January 2025 - January 2026
; Managed a 40-TA team, earning trust and driving mentorship to help 180 students build a solid ECE foothold
- Redesigned ECE labs to deliver meaningful lessons, guiding first-year students through real-world workflows
- Automated course admin via Python scripts, empowering TAs to focus on teaching quality and student mentorship
- One-Instruction Flappy Bird** | github.com/jobitaki/JustOneFlappyBird January 2025
- Co-designed a single-instruction (SUBLEQ) CPU to play Flappy Bird for CMU's Build18 hackathon
- Implemented memory-mapped I/O and VGA graphics in SystemVerilog, deploying to a Spartan-7 FPGA